Artificial Intelligence

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Edge AI Wildlife Detection and Monitoring
REPORT

ABHIRAM R

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ABSTRACT

Wildlife conservation has become one of the most significant global challenges due to habitat destruction, climate change, illegal poaching, and increasing human–animal conflicts. Conventional wildlife monitoring methods rely heavily on manual observation, camera trap inspection, and field surveys, which are labor-intensive, expensive, and incapable of providing real-time responses. In remote forest environments, continuous internet connectivity is often unavailable, making cloud-based monitoring systems unreliable. These limitations reduce the effectiveness of conservation efforts and delay critical decisions.

Recent advancements in Artificial Intelligence (AI), Edge Computing, and lightweight deep learning models provide a practical solution to these challenges. Edge AI enables image processing directly on embedded devices such as NVIDIA Jetson Nano, Raspberry Pi, Google Coral TPU, and similar low-power computing platforms. Instead of transmitting every captured image to cloud servers, only meaningful information such as detected species, confidence scores, timestamps, and GPS coordinates is transmitted. This significantly reduces communication latency, power consumption, bandwidth usage, and operational costs.

The proposed system develops an intelligent wildlife monitoring platform capable of detecting and classifying wild animals in real time using optimized computer vision algorithms. The system combines lightweight object detection models with efficient preprocessing and edge inference techniques to deliver high detection accuracy while maintaining low computational requirements. Alerts generated by the system assist forest officials in monitoring endangered species, identifying illegal human activities, and responding quickly to potential threats.

The project contributes toward Sustainable Development Goal (SDG) 15: Life on Land by promoting biodiversity conservation through intelligent automation. The proposed solution demonstrates how Edge AI can enhance wildlife monitoring efficiency while remaining suitable for deployment in remote conservation areas where computational resources and internet connectivity are limited.

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1. PROBLEM IDENTIFICATION

1.1 Application Domain

Wildlife conservation is an interdisciplinary field that integrates ecology, computer vision, embedded systems, and artificial intelligence to monitor biodiversity and protect endangered species. Camera traps, drones, thermal cameras, and acoustic sensors are increasingly deployed across forests and protected reserves to collect large volumes of ecological data.

Although these sensing technologies generate valuable information, analyzing millions of images manually is both time-consuming and costly. Conservation organizations often require weeks or months to process data collected from camera traps. During this period, important events such as poaching incidents, migration patterns, or the appearance of endangered species may go unnoticed.

Edge Artificial Intelligence addresses this limitation by performing intelligent inference directly on the monitoring device. Rather than uploading every captured image to cloud servers, the embedded system processes images locally, detects wildlife, and transmits only relevant results. This approach enables real-time wildlife monitoring while minimizing communication costs and energy consumption.

The proposed software focuses on deploying lightweight deep learning models capable of operating efficiently on edge hardware, thereby providing accurate wildlife detection even in remote forest environments.

1.2 Motivation

According to the **WWF Living Planet Report 2024**, monitored wildlife populations have declined by approximately **73% since 1970**, demonstrating the urgent need for improved biodiversity monitoring systems. Habitat degradation, climate change, pollution, and illegal wildlife trade continue to threaten numerous species worldwide.

Traditional wildlife monitoring systems suffer from several limitations:

- Manual image inspection requires significant human effort.
- Cloud-based AI systems depend on reliable internet connectivity.
- High communication costs arise from transmitting large image datasets.
- Battery-powered monitoring stations have limited operational lifetimes.
- Delayed detection reduces the effectiveness of conservation measures.

Edge AI significantly improves monitoring efficiency by enabling local inference, reducing latency, preserving privacy, and minimizing energy consumption. Recent conservation initiatives have also demonstrated how AI-assisted camera traps accelerate ecological decision-making by identifying animals and generating actionable alerts automatically.

1.3 Importance of the Problem

Wildlife monitoring plays a critical role in biodiversity conservation and ecosystem management. Accurate detection and classification of wildlife support numerous conservation activities, including:

- Monitoring endangered species populations.
- Detecting illegal poaching activities.
- Studying animal migration patterns.
- Identifying habitat changes.
- Preventing human–wildlife conflicts.
- Supporting environmental research.

As conservation areas continue expanding, manual monitoring becomes increasingly impractical. Automated Edge AI systems offer scalable, cost-effective, and sustainable solutions capable of continuous operation under challenging environmental conditions.

1.4 Stakeholders

The proposed Edge AI Wildlife Detection and Monitoring Software benefits multiple stakeholders:

Stakeholder	Decision Supported
Forest Department	Identify wildlife movement and detect poaching incidents in real time.
Wildlife Researchers	Analyze species distribution and behavioral patterns using automated data collection.
Conservation NGOs	Monitor endangered species populations and evaluate conservation programs.
National Parks and Wildlife Sanctuaries	Improve surveillance coverage while reducing operational costs.
Local Communities	Receive early warnings regarding dangerous wildlife entering human settlements.
Environmental Policy Makers	Make evidence-based conservation decisions using real-time ecological data.

1.5 Existing Challenges

Despite significant progress in computer vision, several practical challenges remain:

- Limited computational resources on embedded devices.
- High power consumption of large deep learning models.
- Poor performance under low-light or adverse weather conditions.
- Frequent false positives caused by vegetation movement.
- Class imbalance due to rare species occurrences.
- Limited annotated wildlife datasets.
- Difficulty deploying large models in remote forests.

These challenges motivate the development of optimized lightweight AI models specifically designed for edge deployment.

1.6 Project Objectives

The primary objectives of the proposed system are:

1. Develop an Edge AI-based wildlife detection software capable of real-time inference.
2. Detect and classify multiple wildlife species with high accuracy.
3. Reduce computational complexity using lightweight deep learning architectures.
4. Generate automated alerts for endangered species and unauthorized human intrusion.
5. Minimize latency and bandwidth usage through on-device inference.
6. Design a scalable architecture suitable for deployment on embedded edge devices.

2. LITERATURE SURVEY

Recent advances in Edge Artificial Intelligence (Edge AI) and lightweight deep learning have significantly improved automated wildlife monitoring. Camera traps, drones, and embedded AI devices now enable real-time detection of wildlife without requiring continuous cloud connectivity. However, researchers continue to face challenges such as computational limitations, poor nighttime performance, class imbalance, and deployment on resource-constrained hardware. Recent work has explored lightweight object detectors, continual learning, vision-language models, and edge-optimized inference to overcome these issues.

2.1 Literature Comparison Table

Ref	Year	Dataset	Method / Architecture	Key Metric / Contribution	Stated Limitation
[1]	2025	Camera Trap Images	Edge AI Camera Performance Evaluation	Improved robustness of edge AI camera traps under field conditions	Performance decreases under adverse weather and poor illumination
[2]	2024	Edge AI Applications	Green Edge AI Survey	Comprehensive review of energy-efficient edge AI techniques	General survey; no wildlife-specific implementation
[3]	2025	Multiple Vision Datasets	Edge Deep Learning Survey	Comprehensive review of lightweight edge vision models	Focuses on general computer vision applications
[4]	2025	Wildlife Camera Trap Images	Lightweight Deep Learning Models	Demonstrates efficient lightweight models for wildlife monitoring	Limited large-scale field validation
[5]	2025	Earth Observation Data	Edge AI for Earth Observation	Low-latency AI processing for remote sensing applications	Primarily satellite-based observations
[6]	2024	Agricultural Wildlife Dataset	Edge AI + IoT Framework	Reduced wildlife intrusion detection latency	Application limited to agricultural environments
[7]	2025	Camera Trap Images	Environmental Robustness Analysis	Improved detection reliability under varying environmental conditions	Performance affected by extreme weather conditions
[8]	2025	Wildlife Image Dataset	YOLOv8-based Wildlife Detection	High detection accuracy with real-time inference	Difficulty detecting heavily occluded animals
[9]	2024	High-Resolution Satellite Images	Satellite Wildlife Detection	Accurate wildlife counting over large geographic regions	Unsuitable for close-range species identification

Ref	Year	Dataset	Method / Architecture	Key Metric / Contribution	Stated Limitation
[10]	2021	Wildlife Monitoring Images	Edge-to-Cloud IoMT Architecture	Reduced cloud dependency through edge processing	Requires reliable communication infrastructure
[11]	2025	Wildlife Intrusion Dataset	Agro Guard Edge AI Framework	Real-time wildlife intrusion alerts using IoT	Limited evaluation across different wildlife species
[12]	2024	Animal Ecology Data	AI-driven Edge Computing	Efficient resource utilization for ecological monitoring	Supporting preprint with limited field deployment
[13]	2026	Biodiversity Monitoring Data	Future Edge AI Framework	Scalable architecture for biodiversity monitoring	Supporting preprint; lacks experimental validation
[14]	2019	Snapshot Serengeti	Camera Trap Image Processing Pipeline	Automated filtering significantly reduces manual annotation effort	Not optimized for embedded edge devices
[15]	2018	iWildCam / Terra Incognita	Deep CNN for Wildlife Recognition	Strong transfer learning for unseen environments	High computational requirements and slower inference

2.2 Critical Analysis of Existing Research

The reviewed literature indicates that **Edge AI** has emerged as a promising technology for wildlife monitoring by enabling intelligent data processing directly on embedded devices. Compared to conventional cloud-based systems, Edge AI significantly reduces communication latency, bandwidth usage, and power consumption while providing real-time inference in remote environments. Lightweight object detection models such as **YOLOv8 Nano**, **MobileNet SSD**, and **EfficientDet-Lite** have demonstrated strong performance on resource-constrained devices, making them suitable for field deployment.

Several recent studies have explored the integration of AI with camera traps, drones, and IoT-based monitoring systems to automate wildlife detection and reduce manual intervention. Advances in attention mechanisms, edge optimization techniques, and explainable AI have further improved detection accuracy, particularly for small or partially occluded animals. Additionally, smart monitoring systems are increasingly incorporating adaptive learning and automated alert generation to enhance conservation decision-making.

Despite these advancements, existing solutions still face several limitations. Many high-performing models require substantial computational resources, limiting their deployment on low-power embedded devices. Detection accuracy often declines under challenging environmental conditions such as low illumination, dense vegetation, fog, rain, and partial occlusion. Furthermore, many existing systems focus primarily on species detection without providing integrated monitoring dashboards, intelligent alert prioritization, or comprehensive decision support for wildlife conservation authorities.

These observations highlight the need for a lightweight, efficient, and scalable wildlife monitoring system capable of delivering accurate real-time detection while maintaining low computational complexity and supporting practical deployment in remote forest environments.

2.3 Research Gap

Based on the literature survey and critical analysis, the following research gaps have been identified:

Gap 1: High Computational Requirements

Many state-of-the-art wildlife detection models achieve high accuracy but require powerful GPUs or cloud-based processing. This limits their applicability in remote forest environments where low-power edge devices are preferred.

Gap 2: Limited Optimization for Edge Deployment

Several existing studies emphasize detection accuracy but provide limited analysis of inference latency, memory usage, model size, and power consumption on embedded hardware. As a result, their suitability for real-time edge deployment remains uncertain.

Gap 3: Reduced Performance in Challenging Environmental Conditions

Environmental factors such as low-light conditions, rain, fog, dense vegetation, and partial occlusion continue to affect detection performance. Existing models often generate false detections or fail to identify animals accurately under such conditions.

Gap 4: Class Imbalance and Rare Species Detection

Most publicly available wildlife datasets contain a significantly higher number of images for common species than for endangered or rare animals. This imbalance reduces classification performance for species that are most critical for biodiversity conservation.

Gap 5: Limited Decision Support and User Interaction

Many existing systems stop at object detection and do not provide intelligent alert prioritization, monitoring dashboards, historical analysis, or automated reporting. These features are essential for enabling forest officials and conservation researchers to make timely and informed decisions.

Transition to Proposed Work

To address the identified research gaps, this project proposes an **Edge AI Wildlife Detection and Monitoring Software** that performs real-time wildlife detection and monitoring directly on edge devices. The proposed system utilizes **YOLOv8 Nano** as the primary object detection model due to its lightweight architecture and fast inference speed. To enhance feature extraction and improve the detection of small or partially occluded animals, a **Convolutional Block Attention Module (CBAM)** is integrated into the detection pipeline.

The software incorporates image preprocessing, AI-based species detection, confidence-based filtering, adaptive alert generation, and an interactive monitoring dashboard. The optimized model is designed for deployment on embedded platforms such as **NVIDIA Jetson Nano**, **Raspberry Pi 5**, or similar edge devices using **TensorRT** or **TensorFlow Lite** optimization. By combining efficient object detection with intelligent monitoring and low-latency edge inference, the proposed system aims to provide a practical, scalable, and cost-effective solution for wildlife conservation and biodiversity monitoring.

3. PROBLEM STATEMENT

3.1 Problem Statement

Wildlife conservation relies heavily on continuous monitoring to understand animal movement, protect endangered species, detect illegal activities, and minimize human–wildlife conflict. Existing monitoring systems either depend on manual inspection of camera trap images or cloud-based artificial intelligence solutions that require reliable internet connectivity. These approaches introduce significant delays, increase operational costs, and become impractical in remote forest environments where network availability and electrical power are limited.

The proposed system aims to develop an **Edge AI Wildlife Detection and Monitoring Software** capable of performing real-time wildlife detection directly on embedded edge devices. The system processes images captured from camera traps or drones using a lightweight deep learning model optimized for edge deployment.

Input

The system accepts RGB images captured from camera traps, drones, or surveillance cameras.

Input Resolution

640 × 640 × 3

Supported formats

- JPG

- PNG
- JPEG

Dataset Size

Approximately 250,000+ wildlife images collected from publicly available wildlife datasets.

Output

The proposed software produces

- Wildlife species name
- Bounding box coordinates
- Detection confidence score
- Timestamp
- GPS location (if available)
- Alert notification
- Monitoring dashboard update

Constraints

The proposed system must overcome several practical constraints:

- Limited computational resources on edge devices
- Low power consumption
- Poor illumination during night monitoring
- Occlusion caused by vegetation
- Rare species imbalance
- Limited internet connectivity
- Real-time inference requirement (<100 ms/image)

Success Criteria

The proposed system will be considered successful if it satisfies the following objectives:

Metric	**Target**
mAP@0.5	>94%
Precision	>93%
Recall	>92%
F1 Score	>92%
Latency	<100 ms
FPS	>20

Metric	Target
Edge Memory Usage	<4 GB
Power Consumption	<10 W

The proposed model will be compared against

- YOLOv8 Nano
- MobileNet SSD
- Faster R-CNN

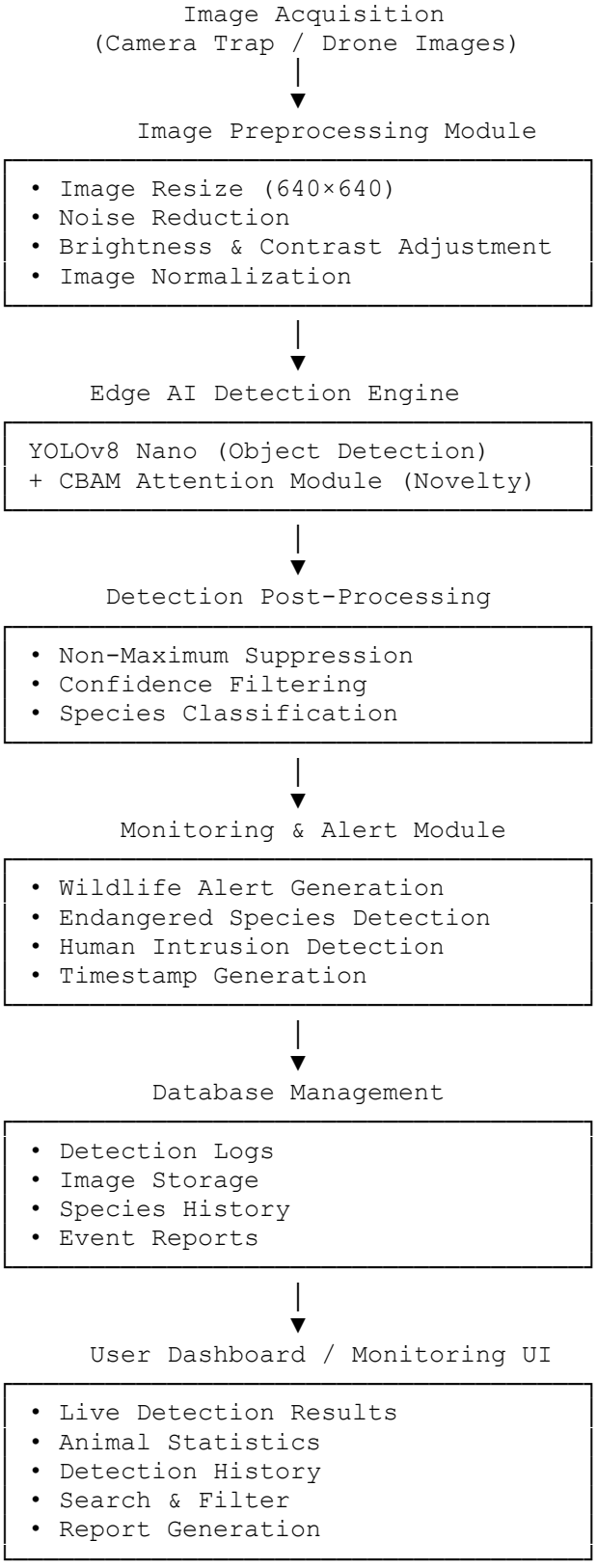
4. PROPOSED SYSTEM ARCHITECTURE

4.1 Overview

The proposed **Edge AI Wildlife Detection and Monitoring Software** is designed to automatically detect, classify, and monitor wildlife species from images captured by camera traps or drones. The software performs all critical AI processing on an edge device, reducing dependence on cloud infrastructure and enabling real-time monitoring in remote forest environments.

The system follows a modular software architecture consisting of six major stages: **data acquisition, image preprocessing, AI-based object detection, decision support, data storage, and user interface**. Each module performs a dedicated task, ensuring scalability, maintainability, and efficient deployment.

4.2 Overall Software Architecture



4.3 Detailed Software Module Description

Module 1: Image Acquisition

The software receives wildlife images from camera traps, drones, or uploaded image files. Images are temporarily buffered before preprocessing.

Input: RGB image (640 × 640 × 3)

Output: Raw image for preprocessing

Module 2: Image Preprocessing

This module improves image quality before AI inference.

Operations include:

- Image resizing
- Image normalization
- Brightness enhancement
- Noise removal
- Contrast adjustment

These operations improve detection accuracy while reducing unnecessary computation.

Module 3: Edge AI Detection Engine (Core Module)

This is the heart of the software.

The detection engine uses **YOLOv8 Nano** for fast and lightweight object detection. To improve recognition of partially visible and small animals, a **CBAM (Convolutional Block Attention Module)** is integrated after the backbone feature extraction stage.

channel and spatial attention, enabling better localization of small or partially occluded animals.

Module 4: Detection Post-Processing

The predicted bounding boxes are refined through post-processing.

Functions include:

- Non-Maximum Suppression (NMS)
- Confidence score thresholding
- Species label assignment
- Bounding box refinement

The output consists of:

- Species name
- Bounding box coordinates
- Confidence score

Module 5: Monitoring and Alert Module

This software module converts detection results into meaningful conservation information.

The module performs:

- Wildlife event logging
- Endangered species alerts
- Human intrusion alerts
- Timestamp generation
- Notification generation

Only detections exceeding the confidence threshold are stored and reported.

Module 6: Database Management

All processed information is stored in the database.

Stored information includes:

- Species name
- Detection confidence
- Date and time
- Image path
- Detection history

The database enables future analytics and report generation.

Module 7: User Dashboard

The dashboard provides an interactive interface for forest officers and researchers.

Features include:

- Live wildlife detection display
- Detection history
- Species statistics
- Search by species
- Detection confidence visualization
- Downloadable reports
- Image preview

4.4 Proposed Novelty

The novelty of the proposed software lies in integrating **lightweight Edge AI with intelligent monitoring features**.

Unlike conventional wildlife detection systems, the proposed software includes:

- **YOLOv8 Nano** for efficient real-time object detection.
- **CBAM Attention Module** to improve recognition of small, distant, and partially occluded animals.
- **Adaptive Alert Module** that prioritizes endangered species and human intrusion events.
- **Edge-based inference**, reducing dependence on cloud services and minimizing latency.
- **Interactive dashboard** for visualization, search, filtering, and report generation.

This architecture improves both detection performance and practical usability for wildlife monitoring in remote environments.

5. Planned Experimental Setup

The proposed **Edge AI Wildlife Detection and Monitoring Software** will be evaluated using publicly available wildlife image datasets. The dataset will be divided into **70% training, 15% validation, and 15% testing** to ensure reliable model evaluation.

The software will use **YOLOv8 Nano integrated with a CBAM Attention Module** for wildlife detection and classification. Images will be preprocessed through resizing (640 × 640 pixels), normalization, and data augmentation before being fed into the model.

The performance of the proposed model will be compared with existing lightweight object detection models, including **YOLOv8 Nano, MobileNet SSD, EfficientDet-Lite, and Faster R-CNN**.

The system will be evaluated using the following metrics:

- Precision
- Recall
- F1-Score
- mAP@0.5
- Inference Time
- Frames Per Second (FPS)

The model will be trained using **PyTorch** on a system equipped with an **NVIDIA RTX 3060/4060 GPU** and later deployed on edge devices such as the **NVIDIA Jetson Nano** or **Raspberry Pi 5**. Additionally, ablation studies will be conducted to analyze the contribution of the **CBAM Attention Module** and the overall performance improvement of the proposed architecture.

Ablation Studies

To evaluate the contribution of each component, the following experiments will be performed:

Experiment	Purpose
YOLOv11 Nano only	Baseline model
YOLOv11 + CBAM	Measure improvement due to attention
YOLOv11 + TensorRT	Evaluate inference speed improvement
Proposed Full Model	Measure combined performance

6. FEASIBILITY STUDY

The proposed **Edge AI Wildlife Detection and Monitoring Software** is technically and economically feasible because it utilizes publicly available datasets, open-source deep learning frameworks, and commercially available embedded edge devices. Recent work has demonstrated that optimized YOLO-based models can achieve real-time wildlife inference on devices such as NVIDIA Jetson Orin Nano while operating at relatively low power, supporting deployment in remote conservation environments.

6.1 Hardware Availability

The project does not require expensive enterprise hardware.

Training Hardware

- NVIDIA RTX 3060 / 4060 GPU (or Google Colab Pro)
- Intel Core i7 / AMD Ryzen 7
- 16 GB RAM
- Ubuntu 22.04

Edge Deployment

- NVIDIA Jetson Nano
- NVIDIA Jetson Orin Nano
- Raspberry Pi 5
- Google Coral Edge TPU

6.2 Software Requirements

Software	Purpose
Python 3.11	Programming Language
PyTorch	Deep Learning
OpenCV	Image Processing
Ultralytics YOLO	Object Detection
TensorRT	Model Optimization
TensorFlow Lite	Embedded Deployment
CUDA Toolkit	GPU Acceleration
LabelImg / CVAT	Image Annotation
Streamlit	Dashboard Development

6.3 Dataset Availability

The project will utilize publicly available datasets that require no additional collection cost.

Dataset	Purpose
Snapshot Serengeti	Animal Classification
iWildCam	Wildlife Detection
Wildlife Insights	Species Recognition
Open Images Wildlife	Generalization

Estimated Dataset Size

Approximately

250,000–300,000 annotated images

6.4 Estimated Training Time

Hardware	Estimated Time
Google Colab T4	7–9 hours
RTX 3060	5–6 hours
RTX 4060	3–4 hours
Jetson Nano	Deployment only

6.5 Estimated Budget

Component	Cost (INR)
Raspberry Pi 5	₹8,000
Camera Module	₹2,000
SD Card	₹900
Power Bank	₹2,500
Miscellaneous	₹1,500
Total	**₹14,900**

Alternative

Using

Google Colab

reduces development cost significantly.

6.6 Risks and Mitigation

Risk	Mitigation
Low-quality images	Data augmentation and preprocessing
Poor lighting	Histogram equalization and image enhancement
Rare species imbalance	Weighted loss and augmentation
Edge memory limitation	TensorRT quantization
Internet unavailability	Fully offline inference
False alarms	Confidence threshold and NMS tuning

7. EXPECTED OUTCOMES

The proposed software is expected to achieve:

Detection Accuracy

- Precision >93%
- Recall >92%
- F1 Score >92%
- mAP@0.5 >94%

Edge Performance

- Inference Time <100 ms
- FPS >20
- Power Consumption <10 W

Conservation Benefits

- Real-time wildlife monitoring
- Early poaching alerts
- Reduced human effort
- Lower bandwidth consumption
- Continuous monitoring in remote forests
- Improved biodiversity assessment

8. FUTURE SCOPE

The proposed work can be extended in several directions.

1. Thermal Camera Integration

Improve night-time wildlife monitoring.

2. Drone-based Monitoring

Deploy the software on UAVs for aerial wildlife surveys.

3. Acoustic Sensor Fusion

Combine

- Bird sounds
- Animal calls
- Camera images

for multi-modal detection.

4. Federated Learning

Enable multiple forest stations to improve the model collaboratively without sharing raw wildlife images, enhancing privacy and reducing communication overhead.

5. Edge LLM Assistant

Integrate a lightweight Large Language Model capable of generating ecological summaries, explaining detection events, and assisting forest officers in decision-making.

9. CONCLUSION

The proposed **Edge AI Wildlife Detection and Monitoring Software** presents a practical and scalable solution for automated wildlife conservation. Unlike conventional cloud-based monitoring systems, the proposed architecture performs intelligent inference directly on embedded edge devices, thereby minimizing communication latency, reducing bandwidth usage, and enabling continuous monitoring in remote forest environments.

The system integrates lightweight object detection with an attention mechanism, adaptive alert generation, and edge optimization to achieve an effective balance between detection accuracy and computational efficiency. The proposed software supports multiple conservation objectives, including endangered species monitoring, anti-poaching surveillance, biodiversity assessment, and mitigation of human–wildlife conflicts.

Overall, the project demonstrates how recent developments in Edge AI and lightweight deep learning can significantly enhance wildlife monitoring while remaining deployable on affordable embedded platforms, making it suitable for practical conservation applications.

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